

Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials and Rational Extension of Various Quantum Mechanical Systems

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Chapter 6

Summary and Prospective

It is now time to summarize the key insights of this thesis and explore potential avenues for future research. This chapter provides a concise overview of the findings discussed in the previous chapters, along with suggestions for future exploration in this field. In our study, we focused primarily on EOPs, which play a crucial role in rationally extended systems. In this thesis, we explore ES potentials in PDM systems, particularly those whose wavefunctions are expressed in terms of EOPs. The role of SUSY and SI symmetries in these systems is thoroughly examined, showing how they contribute to the exact solvability and isospectrality of these potentials. We also explored QES systems, obtaining corresponding zero-energy normalizable solutions. To achieve these results, we employed various mathematical techniques.

This thesis begins by introducing ES and QES systems, along with the significant developments in this area of research. It covers key concepts such as SUSY, SI, and the factorization method, among others, which have played a crucial role in advancing the understanding of these systems. We also discuss the importance and historical background of PDMs and EOPs. Building on these developments, the motivations behind our research are detailed, and a concise overview of the work carried out in this thesis is provided. We have briefly outlined the works that have been carried out in this thesis. Chapter 1 concludes with an outline of the thesis, presenting the chapter-wise plan of the work.

After providing an introductory overview, in the second chapter, we discuss several important mathematical tools essential for our work. As we know, EOPs play a significant role in the wavefunctions associated with rationally extended potentials. We begin by giving a brief introduction to COPs and then demonstrate how EOPs are introduced into the framework. We start with the simplest cases of X_1 -Jacobi and X_1 -Laguerre polynomials.

Following this, we introduce the differential equations associated with these EOPs and discuss the inner product concerning a new weight function. Several examples of these polynomials are provided. Next, we generalize the study of these EOPs for any arbitrary integer $m \geq 1$. We explore their differential equations, weight functions, and the relationship with COPs.

Our study progresses by incorporating the mathematics used in SUSY QM. Here, we delve into the concept of partner potentials and the isospectrality of partner Hamiltonians, examining how their eigenfunctions and eigenvalues are interconnected. We also introduce the concept of a hierarchy of Hamiltonians and, using the well-known concept of SI, show how the eigenvalues and eigenfunctions of this hierarchy of Hamiltonians can be obtained.

Further, we extend our study by exploring a powerful method, the PCT approach, which is crucial for extending conventional potentials to their rationally extended counterparts. This approach is useful for generating both SIPs and non-SIPs, which may be ES, QES, or CES potentials. It also provides a method to find the wavefunctions corresponding to these new potentials. This approach is applicable for both constant mass cases and PDM cases. Finally, we conclude Chapter 2 with a discussion on the symmetry generated by a potential algebra, namely the $SO(2, 1)$ algebra.

We address the question related to the concept of isospectrality. As discussed, there are a limited number of ES potentials, such as the SHO, whose wavefunctions involve Hermite polynomials, the H-atom problem with solutions in terms of Laguerre polynomials, and certain spherically symmetric potentials that generally require Legendre and associated Legendre polynomials as solutions. Some of these potentials have been rationally extended, with their corresponding wavefunctions expressed in terms of EOPs. A fascinating aspect of these conventional potentials and their rationally extended counterparts is that, despite having completely different wavefunctions, they remain strictly isospectral. We demonstrate that the solubility of the differential equations related to these EOPs arises from the underlying SUSY and SI [229]. We start from the differential equation of X_1 -Jacobi and X_1 -Laguerre EOP. To study these differential equations within the framework of SUSY QM, we first express them in the form of eigenvalue equations. This formulation allows us to construct a corresponding Hamiltonian. For these specific polynomials, we can define raising and lowering operators, which play a crucial role in the factorization of the Hamiltonian. The factorization of the Hamiltonian in terms of these operators enables us to introduce a SUSY partner Hamiltonian. In our approach, we define two families of Hamiltonians by introducing a parameter, denoted as k , which can take non-negative integer values ($k = 0, 1, 2, \dots$). We demonstrate that for each value of k , these two families of Hamiltonians are SUSY partners, meaning they are related through SUSY transformations.

Moreover, we show that these Hamiltonians satisfy the condition of SI, a key property in SUSY QM that greatly simplifies the process of finding exact solutions to the eigenvalue problem. By constructing a hierarchy of Hamiltonians, we can systematically determine the eigenvalues of the original Hamiltonian. This hierarchical structure reveals that the solvability of the differential equations governing these polynomials is deeply rooted in the underlying principles of SUSY QM and SI. This insight provides a robust framework for analyzing similar systems.

Furthermore, we extend our analysis to a more generalized case, specifically for the X_m -EOPs, where $m \geq 1$. This generalization broadens the applicability of our findings, demonstrating that the relationship between SUSY, SI, and the solubility of the differential equations holds true across a wide class of exceptional orthogonal polynomials. This comprehensive study highlights the intrinsic connection between the mathematical structure of these polynomials and the physical principles of SUSY QM.

In this study, we extend our investigation within the framework of PDM systems by identifying ES potentials whose corresponding wavefunctions can be expressed in terms of EOPs. We begin by formulating the effective mass kinetic energy operator, which serves as the foundation for constructing a hermitian Hamiltonian for the PDMSE. Our approach adheres to the BenDaniel-Duke assumption, under which the applied potential and the effective potential become identical. We explore two distinct cases of PDMs in this work. The first case involves a well-established mass profile that has been extensively utilized in the literature. The second case introduces a novel PDM [232], which depends on a parameter ν and is subject to specific constraints. Using the PCT approach in the context of PDM systems, we first apply it to the known PDM previously used by Midya et al. [231]. We successfully derive ES potentials that are parameterized by an integer m . Additionally, we determine the corresponding energy eigenvalues and wavefunctions for these new potentials, which also depend on the integer parameter m . For a specific value of $m = 1$, our results reproduce those obtained by Midya et al. [231]. For this case, we have plotted the potential, the probability distribution of the initial wavefunctions, and the PDM function for different values of m .

Subsequently, we repeat the analysis for the second PDM class, where the ES potentials are now functions of both the integer parameter m and the restricted parameter ν . We derive the corresponding wavefunctions and energy eigenvalues, which exhibit dependence on these two parameters. Similar to the first case, we present plots of the probability distributions of some initial wavefunctions, the potential, and the PDM function for different values of m and ν . Finally, we apply the SUSY technique within the PDM framework for both cases. We derive the superpotential and the associated partner potentials, which are SIPs. Notably, in each case, one of the partner potentials matches our previously identified

ES potentials, indicating that these ES potentials are indeed SIPs. This connection underscores the fact that the exact solvability of these potentials is inherently linked to their SI.

In the fifth chapter, we initially introduce the conventional truncated Calogero-Sutherland model. We then extended this model by incorporating rational extensions and provided the corresponding Hamiltonian for the modified system. To solve the Schrödinger equation associated with this extended Hamiltonian, we employed a strategic transformation of the wavefunction. Specifically, we expressed the wavefunction as a product of two functions, one that satisfies the Laplace equation and another that obeys a radial equation. The solution of this radial equation was achieved using the PCT approach, which led to the derivation of an energy expression. To construct QES potentials, we adopted the approach developed by Bagchi et al. [227]. Through this method, we derived a new energy expression that was distinct from the initial one obtained via PCT. By equating this newly found expression to the equation associated with the Casimir operator, we uncovered three distinct conditions. These conditions correspond to three different restrictions imposed on the guiding function related to the $SO(2, 1)$ potential algebra. This scenario is reminiscent of a previous study where a similar situation arose. However, in that earlier work, the wavefunction was normalizable at zero energy only in one case.

In contrast, our investigation yielded a richer set of results [234]. By setting the energy parameter $E = 0$ and employing a carefully chosen mapping function, we focused on the first class of solutions for the guiding function. This led to the discovery of a class of QES potentials. For these potentials, we derived the corresponding wavefunctions and, by imposing certain restrictions on the parameters, we demonstrated that the wavefunctions represent normalizable zero-energy states.

We then extended this analysis to the other two classes of solutions for the guiding function. By following a similar procedure, constructing the QES potentials, and applying suitable restrictions on the parameters involved in these potentials, we found that the wavefunctions corresponding to $E = 0$ in these two additional cases were also normalizable. This comprehensive analysis provided us with three distinct QES potentials, each associated with a normalizable zero-energy wavefunction. To further illustrate our findings, we plotted the wavefunctions and their corresponding probability distributions for each of the three cases, considering different values of the parameters that appear in the potentials.

The work presented in this thesis opens several avenues for future research, particularly in the realm of rationally extended ES potentials. While this work has focused on the rational Jacobi and Laguerre polynomials, there is another very well known family of orthogonal polynomials, i.e., the exceptional Hermite orthogonal polynomials, that may

also exhibit interesting properties within the SUSY and SI frameworks [267]. Investigating this could yield further generalizations of the results obtained here.

A significant number of second-order differential equations in mathematical physics, particularly those derivable from a master function, inherently possess the properties of SUSY and SI. These symmetries have allowed for the derivation of operator product formulas for the polynomial solutions of these equations, offering a powerful framework for understanding and solving them. Building on this foundation, a promising direction for future research lies in extending the master condition to include EOPs [268]. By broadening the scope of the master function to accommodate EOPs, we can potentially uncover new classes of ES and QES models.

In this thesis, we have utilized the $SO(2, 1)$ algebra of potentials to identify QES potentials associated with normalizable zero-energy state solutions. This approach has proven to be highly effective in uncovering new QES potentials and understanding their underlying structure. A compelling direction for future research is to explore the application of the $SO(2, 1)$ algebra within the framework of PDM systems. By integrating the $SO(2, 1)$ algebra with PDM, we can potentially discover new classes of QES potentials that exhibit unique properties and behaviors. This exploration could lead to a deeper understanding of the interplay between algebraic symmetries and variable mass systems, offering insights into more complex and realistic physical models.

These future directions not only aim to deepen our understanding of QM but also to expand the applicability of the techniques developed in this thesis to broader areas of physics.